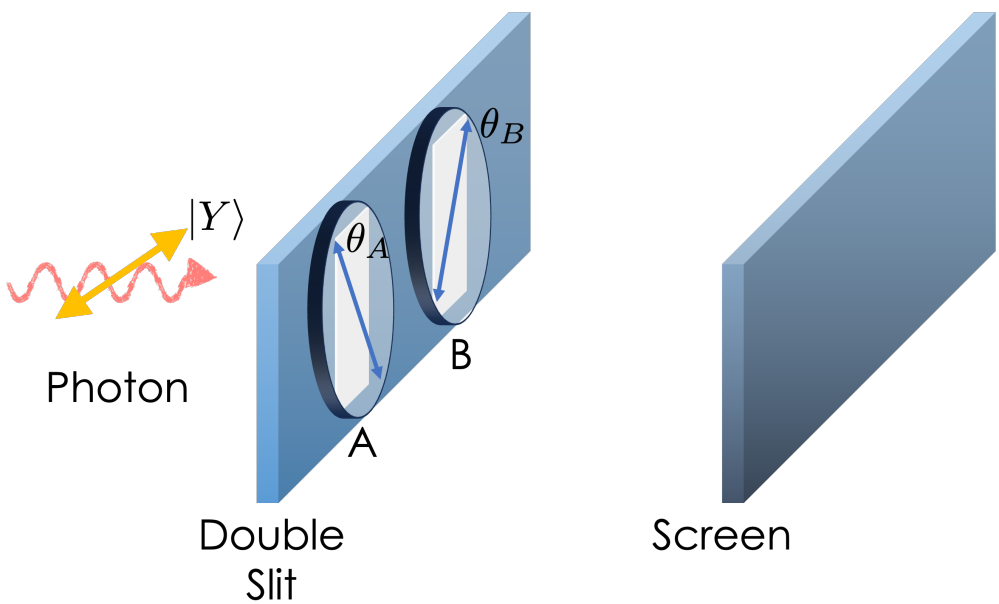
Homework 4

Foundations of Quantum Optics

OPTI 544, Spring 2026

1. **Quantum Erasure**: Consider a single photon going through a double slit as shown in the figure below:



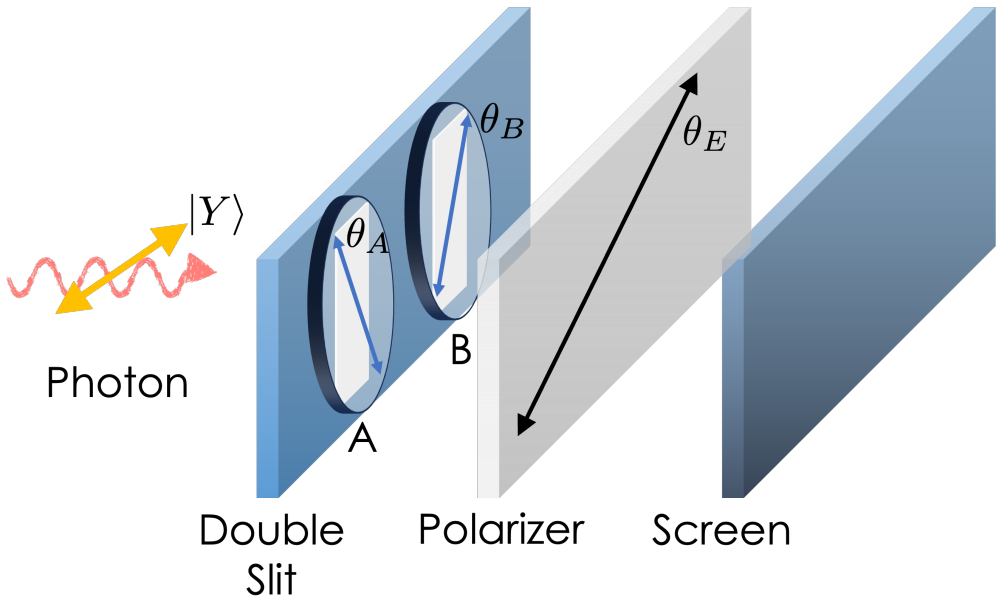
The incoming photon has a polarization state given by $|Y\rangle$. There are two linear polarizers in front of slits A and B oriented along $\theta_A = -\theta_B = \theta$ such that their corresponding operators are given by $\hat{P}_A = |\theta_A\rangle\langle\theta_A|$ and $\hat{P}_B = |\theta_B\rangle\langle\theta_B|$, with $|\theta_A\rangle = \cos\theta\,|X\rangle + \sin\theta\,|Y\rangle$ and $|\theta_B\rangle = \cos\theta\,|X\rangle - \sin\theta\,|Y\rangle$.

(a) What is the state $|\Psi\rangle$ of the photon once it passes through the double slit?

(b) What is the probability $\mathcal{P}_k = |\langle x_k|\Psi\rangle|^2$ of finding the photon at a position x_k on the screen?

(c) How does the probability change as a function of θ? What is the probability for $\theta = 0$? $\theta = \pi/4$? Is there an interference pattern detected on the screen in each of these cases?

We now place an eraser in the path of the photon after the double slit as shown in the figure below:

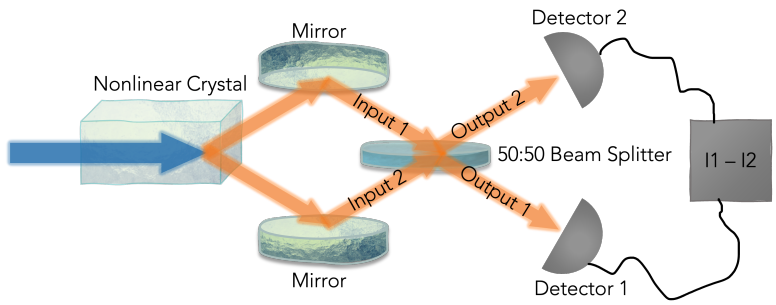


(d) What is the state of the photon once it passes through the polarizer after the double slit?

(e) Consider the case where $\theta = \pi/4$ and $\theta_E = \pi/4$, is there an interference pattern detected on the screen?

(f) For $\theta = \pi/4$ and $\theta_E = 0$, is there an interference pattern detected on the screen?

2. **Hong-Ou-Mandel setup**: Consider the setup shown in the figure below:



A classical field incident on the nonlinear crystal leads to generation of entangled photon pairs. These entangled photons are then incident on a 50:50 beam splitter, with the two output beams being detected on two detectors D1 and D2. Consider the state of the two photons exiting the nonlinear crystal as follows: (i) $|1\rangle_1 |1\rangle_2$, (ii) $\frac{1}{\sqrt{2}}(|1\rangle_1 |0\rangle_2 + |0\rangle_1 |1\rangle_2)$, and (iii) $\frac{1}{\sqrt{2}}(|1\rangle_1 |0\rangle_2 - |0\rangle_1 |1\rangle_2)$, where the subscripts refer to the two input channels (1 and 2). Find the expected number of photons in output 1 (I_1), output 2 (I_2) and the difference ($I_1 - I_2$).

3. **Atomic structure**: Using the Bohr model for the atom, determine the following dimensionless quantities for the ground state electron ($n = 1$) in a Hydrogen atom ($Z = 1$):

   (a) The ratio r_0/λ_0, where r_0 is the Bohr radius, and $\lambda_0 = 2\pi\hbar c/E_0$ is the resonant wavelength associated with the energy of the electron in $Z = 1$. Express this in terms of the fine structure constant $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$.

   (b) The ratio of v_0/c, where v_0 is the velocity of the electron in the ground state and c is the speed of light. Express this in terms of the fine structure constant $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$.

   (c) Show that the ratio g/ω_0, where $g = \frac{\mathbf{d}\cdot\mathbf{E}}{\hbar} \sim \frac{er_0}{\hbar}\sqrt{\frac{\hbar\omega_0}{2\epsilon_0 V}}$ is the vacuum Rabi coupling,

scales as:

$$\frac{g}{\omega_0} \sim \frac{\alpha^{3/2}}{(V/\lambda_0^3)^{1/2}}.$$

The fact that the fine structure constant α is a small number limits the strength of the single photon-single atom interaction. One way to boost this is by reducing the mode volume of the field, as appears in the denominator.